

# Continuity and the Intermediate Value Theorem

ANSWER KEY



## The three-part continuity test

- 1 State the three conditions that must all hold for  $f$  to be continuous at  $x = a$ .  
(1)  $f(a)$  is defined; (2)  $\lim_{x \rightarrow a} f(x)$  exists; (3)  $\lim_{x \rightarrow a} f(x) = f(a)$ .
- 2 Determine whether  $f(x) = \frac{x^2 - 4}{x - 2}$  (for  $x \neq 2$ , with  $f(2) = 4$  defined separately) is continuous at  $x = 2$ .  
 $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 2) = 4$ . Since  $f(2) = 4$  matches, all three conditions hold:  $f$  is continuous at  $x = 2$ .

## Finding values that ensure continuity

- 3 Find  $k$  so that  $f(x) = kx^2$  for  $x \leq 2$ , and  $f(x) = 5x - 2$  for  $x > 2$ , is continuous at  $x = 2$ .  
Need  $4k = 5(2) - 2 = 8$ :  $k = 2$ .
- 4 Find  $a$  and  $b$  so that  $f(x) = 2$  for  $x \leq -1$ ,  $f(x) = ax + b$  for  $-1 < x < 2$ , and  $f(x) = 6$  for  $x \geq 2$ , is continuous everywhere.  
At  $x = -1$ :  $-a + b = 2$ . At  $x = 2$ :  $2a + b = 6$ . Subtracting:  $3a = 4$ , so  $a = \frac{4}{3}$ , and  $b = 2 + a = \frac{10}{3}$ .

## Types of discontinuity

- 5 Classify the discontinuity of each function at the indicated point:
  - a  $f(x) = \frac{x+3}{x^2-9}$  at  $x = -3$  Removable —  $\frac{x+3}{(x-3)(x+3)}$  simplifies to  $\frac{1}{x-3}$  for  $x \neq -3$ .
  - b  $f(x) = \lfloor x \rfloor$  at  $x = 2$  Jump — the floor function steps from 1 to 2.
  - c  $f(x) = \frac{1}{(x-1)^2}$  at  $x = 1$  Infinite —  $f \rightarrow +\infty$  from both sides.

## The Intermediate Value Theorem

- 6 Show that  $f(x) = x^3 - 3x + 1$  has a root in  $(0, 1)$ , using the IVT.  
 $f(0) = 1 > 0$  and  $f(1) = 1 - 3 + 1 = -1 < 0$ . Since  $f$  is continuous (a polynomial) and changes sign on  $[0, 1]$ , the IVT guarantees a root in  $(0, 1)$ .
- 7 A continuous function  $g$  satisfies  $g(1) = -3$  and  $g(5) = 7$ . Explain why  $g$  must take the value 0 somewhere in  $[1, 5]$ , and state the theorem used.  
By the IVT, since  $g$  is continuous on  $[1, 5]$  and 0 lies between  $g(1) = -3$  and  $g(5) = 7$ , there must exist some  $c \in [1, 5]$  with  $g(c) = 0$ .

- 8 A cup of coffee's temperature (in °C) is a continuous function of time, starting at 90°C and cooling to 25°C over 30 minutes. Explain, using the IVT, why the coffee must have been exactly 50°C at some point. Temperature is a continuous function of time; since 50 lies between the starting value 90 and ending value 25, the IVT guarantees the temperature equaled 50°C at some instant during the 30 minutes.
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### Continuity on an interval

- 9 Determine the largest interval(s) on which  $f(x) = \sqrt{9 - x^2}$  is continuous.  
The expression under the square root must be non-negative:  $9 - x^2 \geq 0 \Rightarrow -3 \leq x \leq 3$ .  $f$  is continuous on the closed interval  $[-3, 3]$ .
- 10 Determine all values of  $x$  where  $f(x) = \frac{x+1}{x^2-x-6}$  is discontinuous, and classify each discontinuity.  
Factor denominator:  $x^2 - x - 6 = (x - 3)(x + 2)$ . Discontinuities at  $x = 3$  and  $x = -2$ . Since the numerator  $x + 1$  shares no common factor, both are infinite (vertical asymptote) discontinuities, not removable.
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### Using the IVT to approximate roots

- 11 Show that  $f(x) = \cos x - x$  has a root in  $(0, 1)$ , then use the IVT with the midpoint to narrow the interval to a length-0.5 subinterval.  
 $f(0) = \cos 0 - 0 = 1 > 0$ .  $f(1) = \cos 1 - 1 \approx 0.540 - 1 = -0.460 < 0$ . Sign change confirms a root exists in  $(0, 1)$ . Midpoint  $x = 0.5$ :  $f(0.5) = \cos(0.5) - 0.5 \approx 0.878 - 0.5 = 0.378 > 0$ . Since  $f(0.5) > 0$  and  $f(1) < 0$ , the root lies in  $(0.5, 1)$ .